Progressivity and redistribution of wealth taxation in Spain

A microsimulation analysis using tax register data

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Session: Wealth and fiscal and monetary policy

Presented by the discussant: **Max Ghenis** (PolicyEngine)

Author response: Olga Cantó & Francisco García-Rodríguez

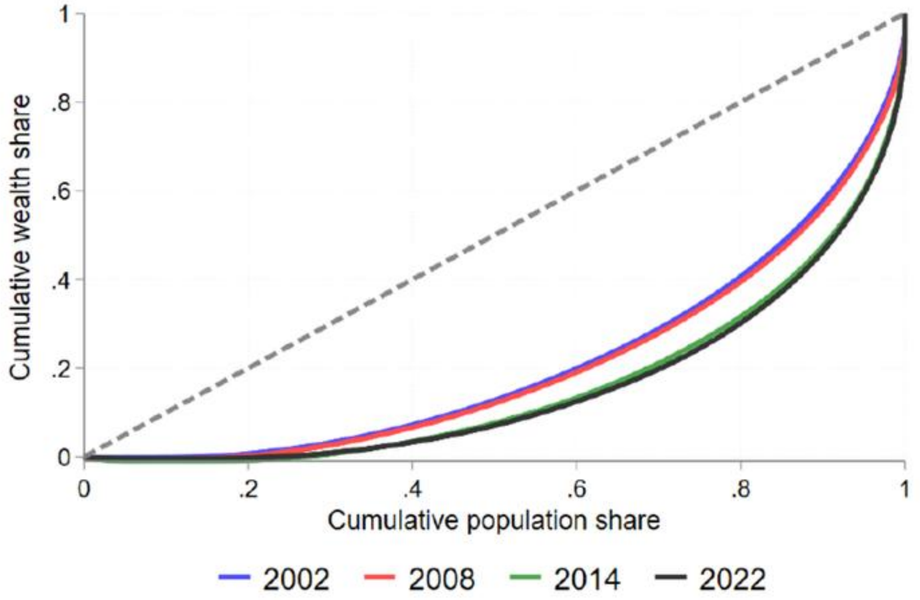
Wealth concentration in Spain has been rising

And most countries have walked away from the tax that targets it

Survey of Household Finances (EFF): the **top 1% share** of net wealth rose from **13.3% (2002) to ~21% (2022)**; the bottom 50% fell from 12.8% to 7.1%.

Meanwhile the annual net wealth tax has nearly vanished: **12 OECD countries in 1990 → 3 by 2023** (Spain, Norway, Switzerland). France retreated to real estate only (ISF → IFI, 2018).

Spain is the natural laboratory: progressive rates, broad legal base, and **17 regional regimes** that can — and do — hollow the tax out.



Lorenz curves of equivalized net wealth (EFF, 2002–2022) — their Figure 1

The Spanish wealth tax: progressive on paper, hollowed out in practice

Impuesto sobre el Patrimonio (IP) — state schedule, regional control

Statutory design (2022 baseline)

- Net wealth above a **€700,000 general exemption** + €300,000 for the main residence
- Progressive state schedule, **0.2% → 3.5%**
- Joint IRPF–IP cap: combined liability ≤ 60% of the income-tax base, with a 20% minimum wealth-tax payment

A tax with a strange life

Introduced 1977 as "temporary"; permanent 1991; suspended 2008; reinstated 2011 — formally temporary, renewed every year since.

Regional autonomy is the story

- Since 2011 regions set rates, thresholds, and rebates — **Madrid and Andalusia: 100% rebate in 2022**
- Catalonia, Extremadura, Valencia keep high effective burdens → horizontal inequity and fiscal competition

The result, in their 2022 data:

170,404 effective taxpayers

of 44.9M individuals (0.4%) · €1.13bn revenue · zero measurable Gini reduction

Scope note: the 2022 state "solidarity" surcharge on large fortunes (ITSGF) is deliberately excluded from the baseline (their footnote 2) — it returns in the discussion.

This paper: what would a different design deliver?

Hold the 2022 population and wealth fixed; swap the tax law; compare

Questions

- How would alternative designs change progressivity and revenue?
- How much redistribution could the revenue buy?
- Which designs bend long-run wealth concentration?
- Who would actually pay, under each?

Contributions

- First wealth-tax design comparison on Spanish **administrative register data**
- Decomposition of redistributive change: **level (incidence) vs distance (progressivity)**
- Long-run (25–100 year) cumulative framework
- Taxpayer profiles under each design (logit)

The comparison set is unusually rich: three real foreign systems (Norway, Switzerland, France), three internal reforms, and Piketty's (2014) normative schedule — all run on the same microdata.

Data: the tax agency's Household Panel

Administrative records for filers and non-filers, compiled by AEAT and provided through the Instituto de Estudios Fiscales

Panel de Hogares (PH), 2022 wave

- **~950,000 households · 2.7M individuals** sampled; weighted to 17.7M households / 44.9M individuals
- Links income-tax returns, wealth-tax declarations, and third-party reports (employers, banks) — covers **non-filers** too
- Panel structure from 2016; used here as a 2022 cross-section

The valuation fix

Real estate is recorded at cadastral value → rescaled to market value with an EFF-based ratio of **×3.08** (uniform across properties).

2022 headline statistics (their Table A.1)

Mean / median net wealth, all individuals	€160,540 / €60,372
Wealth-tax filers (weighted)	170,408
Mean / median net wealth among filers	€2.05M / €1.43M

Limits the authors flag

- Business assets recorded only for wealth-tax filers
- Liabilities not decomposed (EFF used as complement)
- Cadastral values lag markets, unevenly across areas

Eight designs, one fixed population

Same 2022 microdata under each design; no behavioral response — design effects only

Design	What it is	Entry point	Rates
IP (baseline)	Spanish wealth tax as legislated for 2022, all 17 regional regimes	€700k + €300k residence	0.2–3.5% progressive
Sim. NO	Norwegian design: broad base, residence at 25% of market value	~€160k	flat 1% (0.3% + 0.7%)
Sim. CH	Swiss design (Canton of Zurich): real estate at 70% of market value	cantonal thresholds	progressive, low rates
Sim. FR	French IFI: real-estate wealth only	€1.3M trigger	0.5–1.5% progressive
Sim. 1	Spanish IP with the general exemption cut to €300k	€300k	IP schedule
Sim. 2	Spanish IP without regional deductions and rebates	€700k	IP schedule
Sim. 3	Sim. 1 + Sim. 2 combined	€300k	IP schedule
Sim. Piketty	Paper's Piketty-inspired variant; base details incomplete	€200k	0.5% / 1% / 2%

The paper drops Spain's joint IRPF–IP cap in the foreign designs; France's IFI has its own 75% income plafonnement. Sources: their Table 1 (p.11) and Table A.2 (p.40).

Spain IP first: the steepest schedule on paper

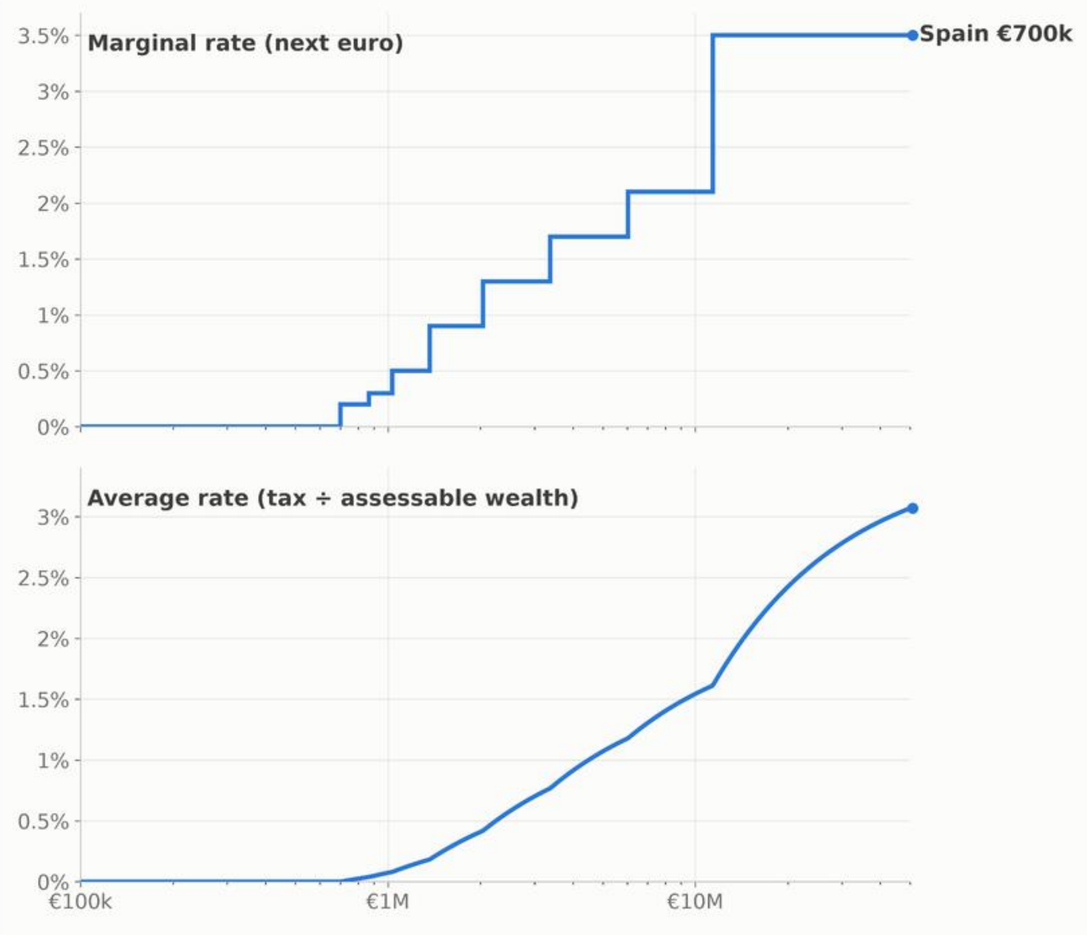
Schedule 1 of 6 — discussant's reference; links to the paper's variants are stated explicitly

Base and unit
State IP schedule on net wealth above €700k. This comparison strips the €300k residence relief, regional rules, and the income cap.

Rates
0.2% → 3.5% over eight brackets — the steepest top rate of the designs you are about to see.

Link to the paper
This is the state-default reference for Sim. 2; the paper leaves its treatment of regional scales and thresholds unresolved. Baseline IP varies across regions.

Shape
Late but fierce: on paper Spain out-taxes everyone at the top. The paper is about how little of this survives exemptions, rebates, and the cap.



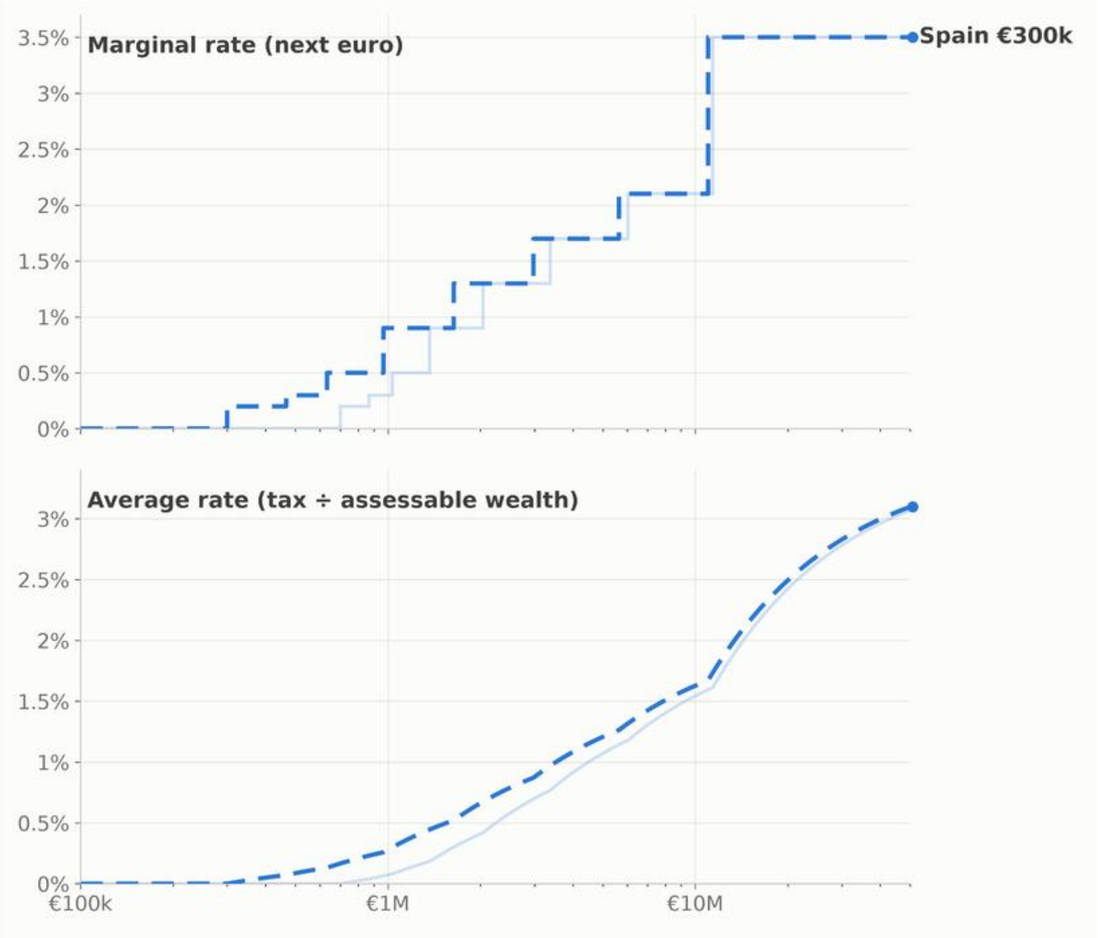
The paper's own reform: pull the threshold to €300k

Schedule 2 of 6 — the shared Spanish rate-and-threshold skeleton

Base and unit
Same state schedule with a €300k exemption: a reference skeleton for Sims 1 and 3, whose design-specific regional parameters are not fully published.

Rates
Same 0.2% → 3.5% brackets; every step arrives €400k of wealth earlier.

Shape
The dashed twin, shifted left. Effective taxpayers rise from 170,404 to 1.8M (Sim. 1) or 4.6M (Sim. 3), but this chart is not an exact recovery of either design.



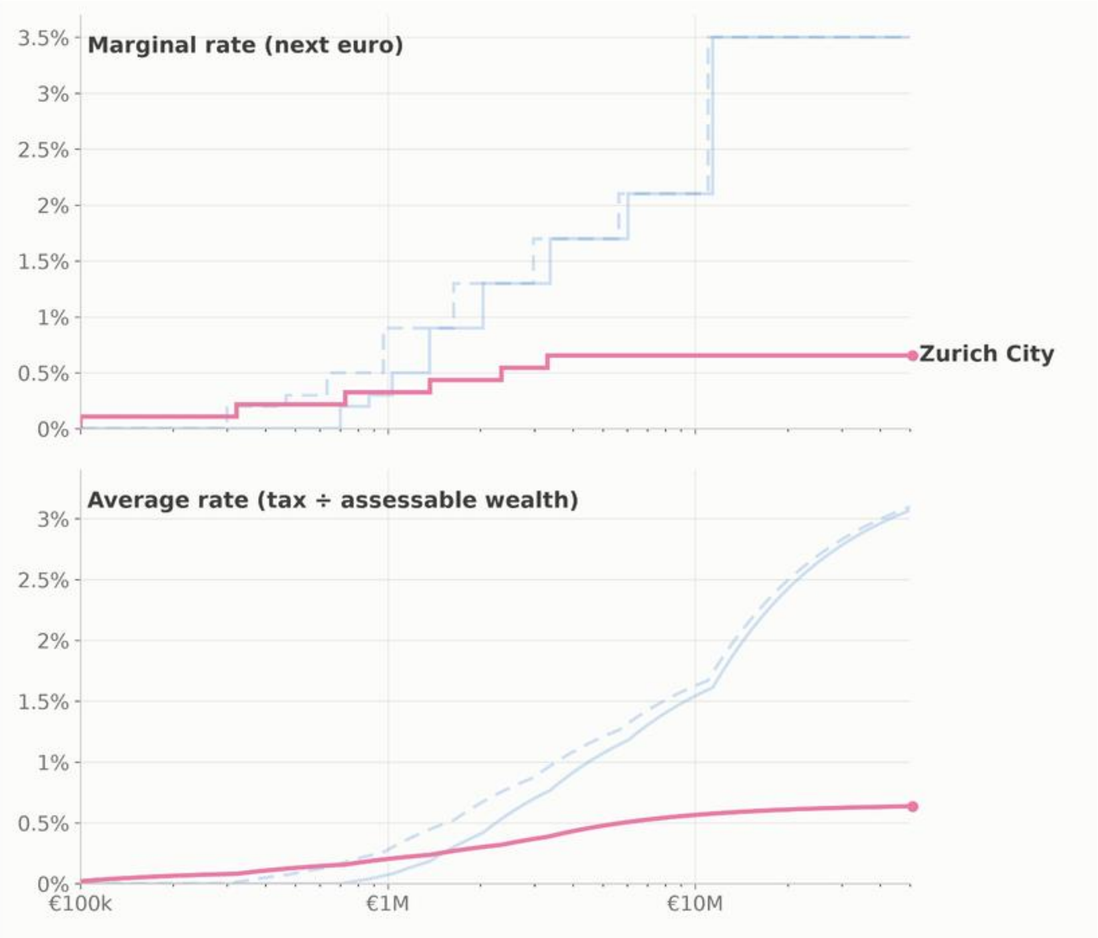
Canton of Zurich: the broad, gentle reference

Schedule 3 of 6 — illustrative Zurich-city benchmark; paper omits vintage and multiplier

Base and unit
Zurich §47 single tariff on net wealth; spouses are jointly taxed and minors attributed. Shown allowance: CHF 81k (2026-indexed table).

Rates
2026-indexed tariff 0–3‰ ×2.18 using canton (99%) + City of Zurich (119%) factors documented for 2022–23 → top ≈0.65%. This is a mixed-vintage illustration.

Shape
Bites almost immediately, climbs slowly, never gets steep — a transparent Zurich-city reference, not a recovered paper parameter.



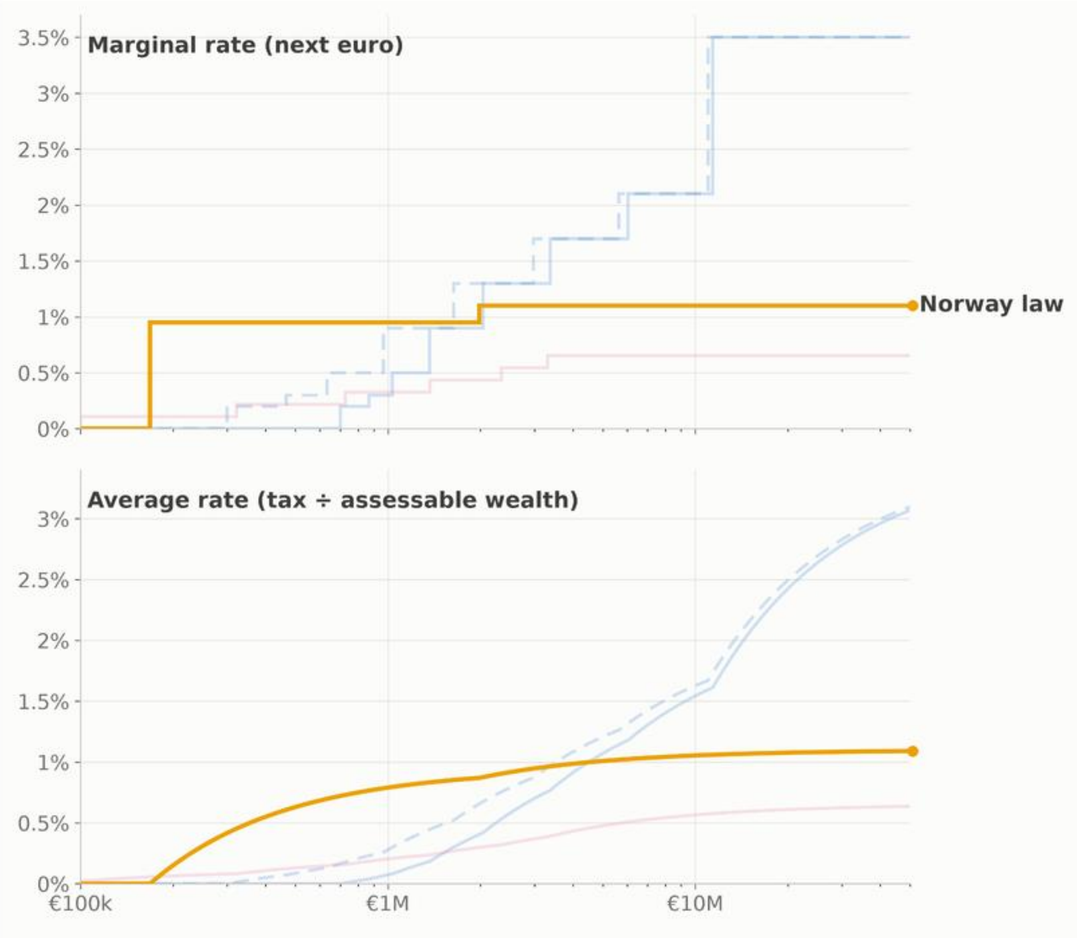
Norway: broad, nearly flat — and not quite the paper's line

Schedule 4 of 6 — 2022 statute; the paper's Sim. NO is flat 1%

Base and unit
All net wealth above NOK 1.7M (≈€168k; doubled for couples). Primary residence is valued at 25% through NOK 10M, then 50%; the paper's Sim. NO keeps 25% throughout.

Rates
0.95% flat in 2022, 1.10% above NOK 20M (≈€2M) — barely progressive by design (the paper's Sim. NO stylizes this as flat 1%).

Shape
A near-flat wall from ~€170k up: mass participation, low rates. Raised ≈0.5% of GDP in 2022.



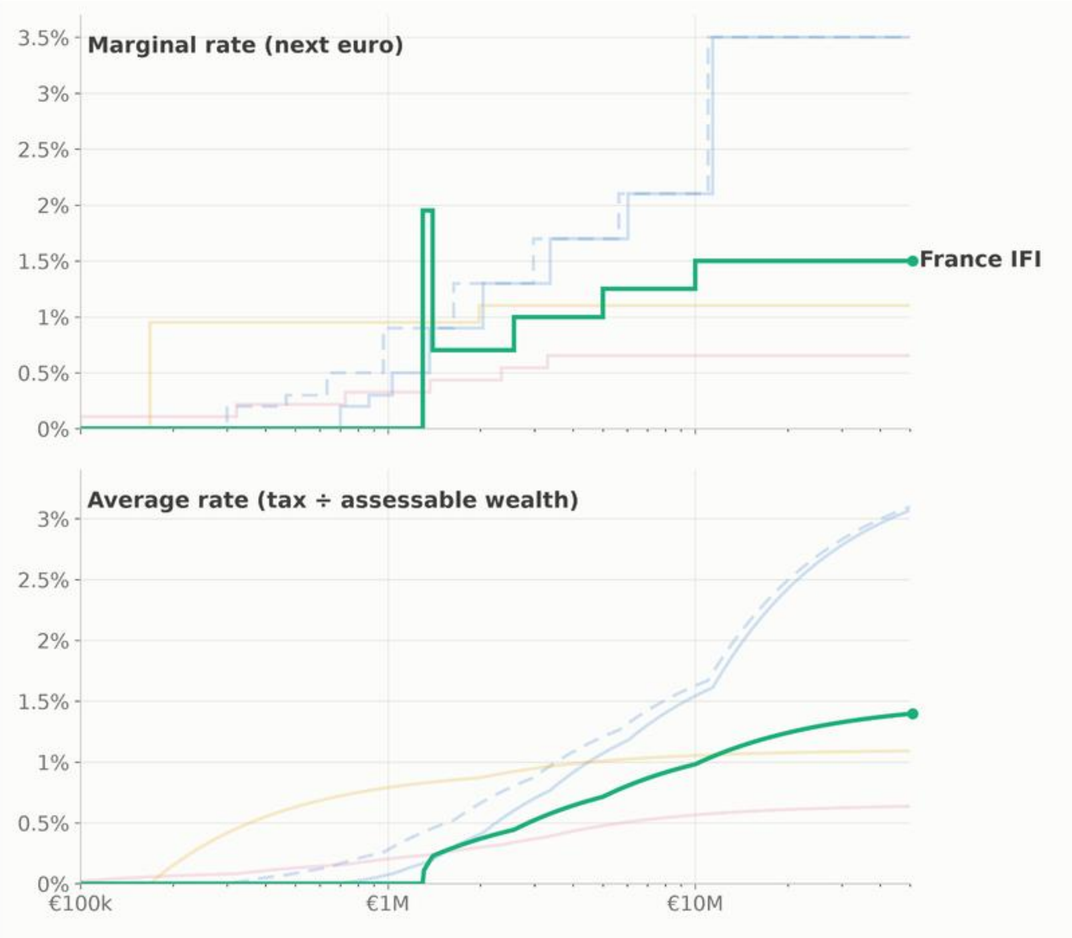
France: narrow base, sharp entry

Schedule 5 of 6 — statutory IFI reference used by the paper's Sim. FR

Base and unit
Real-estate wealth only (the IFI, successor to the ISF since 2018) — financial assets exempt. Trigger at €1.3M.

Rates
0.5–1.5% schedule counted from €800k once over the trigger — so ≈€1,250 falls due just above €1.3M, smoothed by a décote to €1.4M.

Shape
Nothing, then a cliff, then modest progression — and only on property.



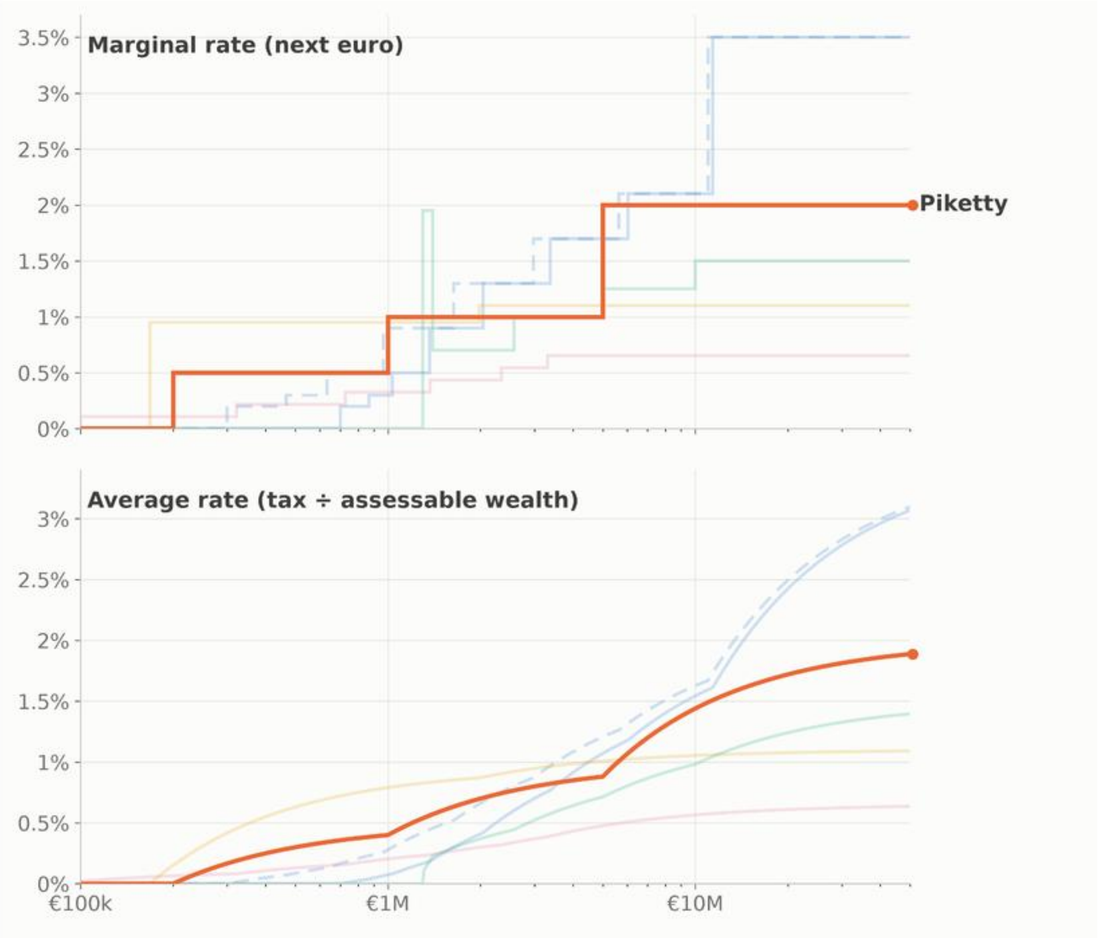
Piketty: the normative benchmark

Schedule 6 of 6 — the paper's normative benchmark, not enacted law

Base and unit
The paper's benchmark begins at €200k. Its published parameter table does not specify the valuation base or whether an income cap carries over.

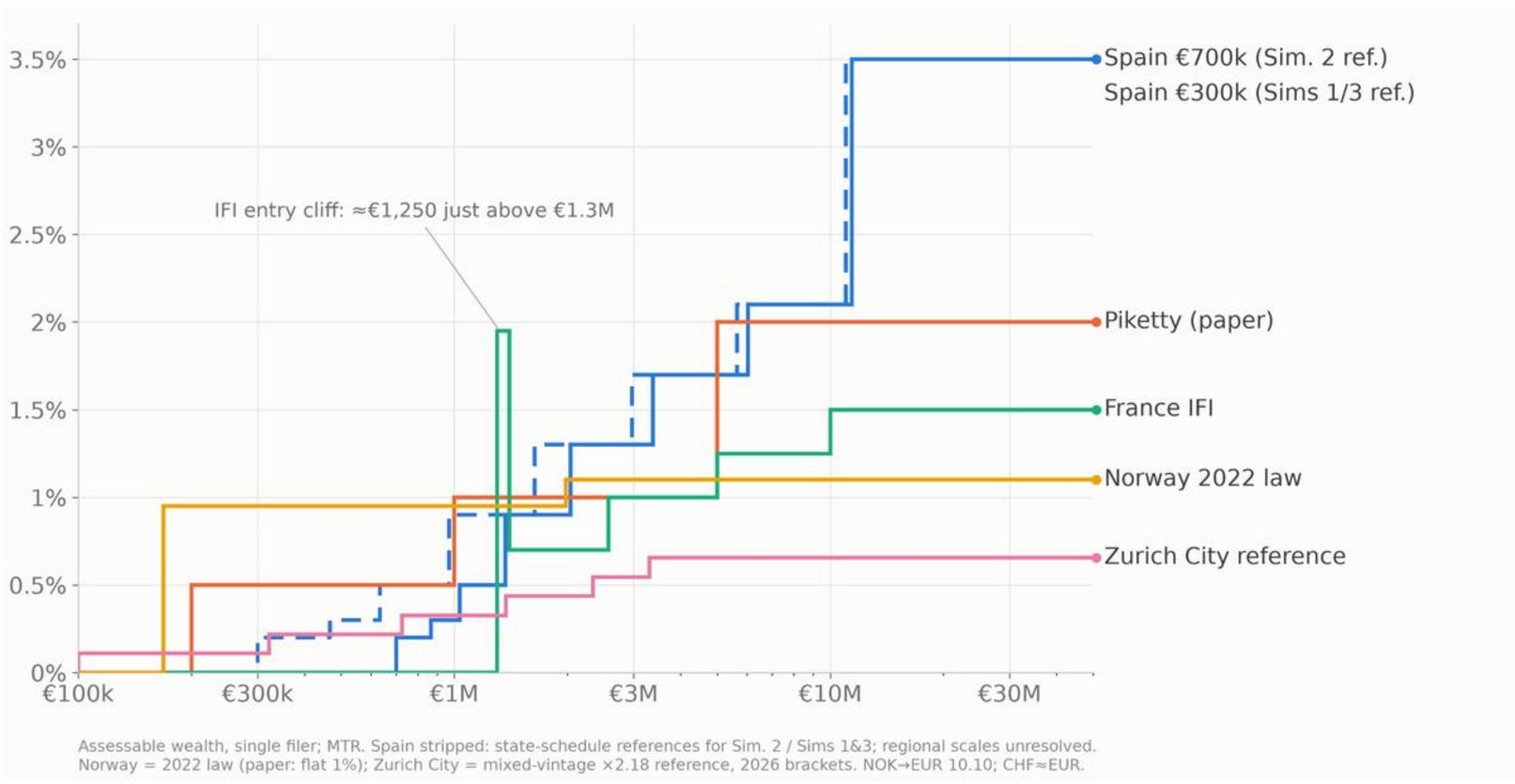
Rates
0.5% / 1% / 2% above €200k / €1M / €5M (the book also floats 0.1% below €200k and 5–10% at the very top).

Shape
Smoothly progressive from modest wealth up. Note how it shadows the dashed €300k Spanish line through the middle of the distribution.



Six reference schedules behind eight designs

Statute/paper benchmarks; Spain stripped of abatements; assumptions labeled



Zurich enters first; Norway follows with a low, nearly flat schedule; un-abated Spanish state IP reaches the steepest top rate (3.5%). Valuation, exemptions, regional rules, and caps turn these schedule shapes into the paper's very different effective liabilities.

How they measure it

Standard fiscal-incidence toolkit, plus a decomposition and a long-run extension

Progressivity & redistribution

- **Kakwani** $K = C_t - G_{pre}$: is the burden more concentrated than wealth?
- **Suits** index (inequality-level-free cousin)
- **Reynolds–Smolensky** $R–S = G_{pre} - G_{post} = t/(1–t)\cdot K - D$
- Measured on net wealth *and* equivalized market / disposable income

Level vs distance

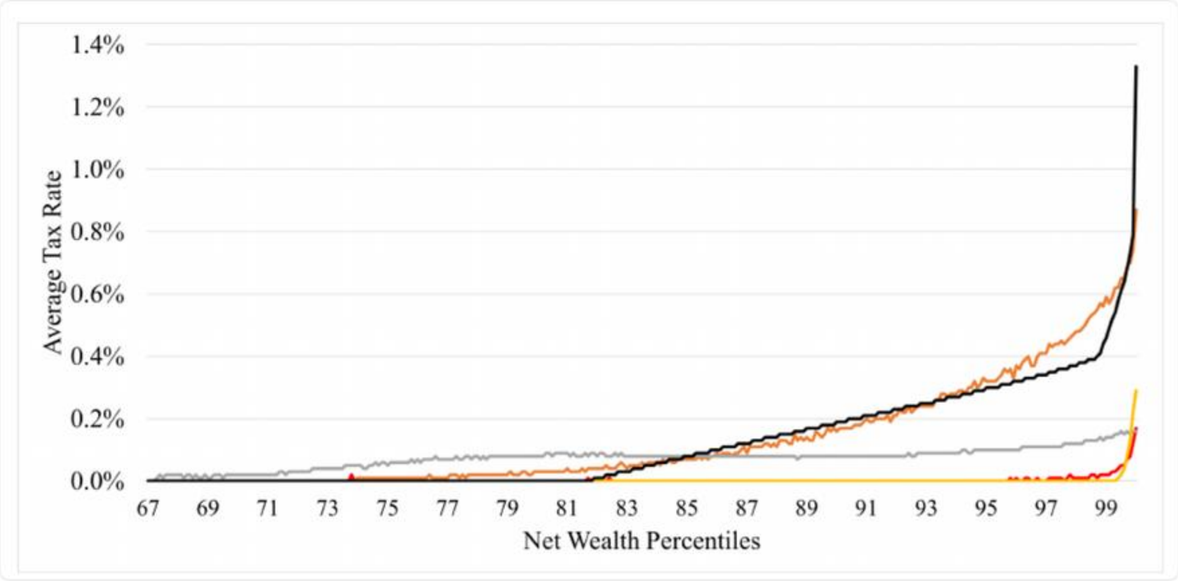
Changes in R–S across designs split into a **level effect** (more incidence — the tax simply collects more) and a **distance effect** (more structural progressivity), following Díaz et al. (2010, 2011).

Long run

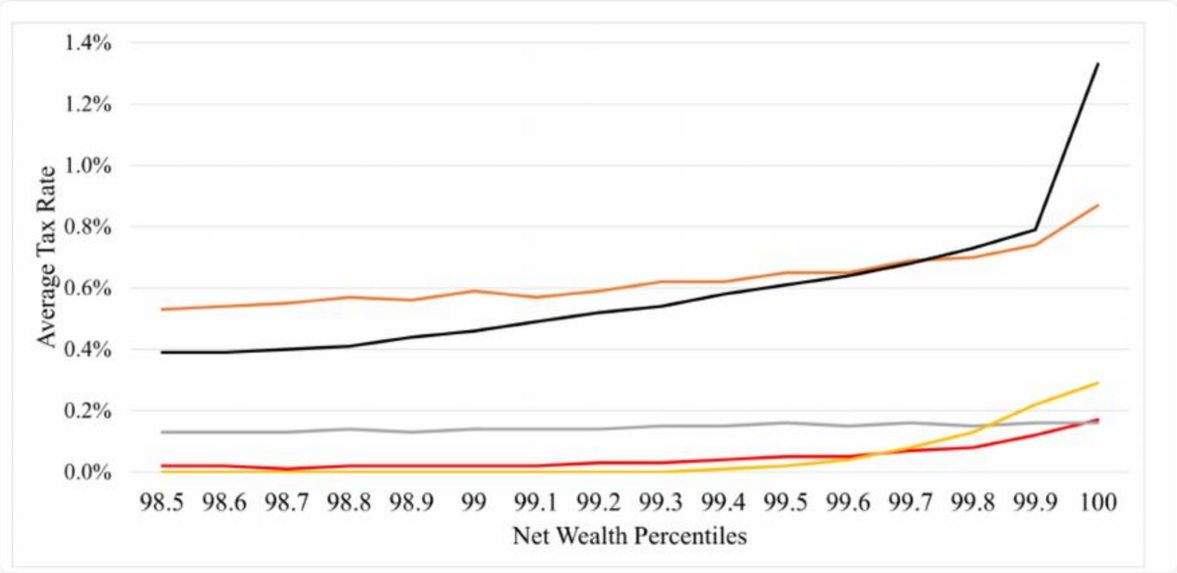
Wealth is a stock: project each person's wealth forward at a constant growth rate $g \in \{1\%, 2\%, 3\%\}$ with and without the annual tax, for T = 25–100 years (Durán-Cabré & Esteller-Moré framework). Compounding turns tiny annual rates into large cumulative wedges.

Result 1 · Who faces the tax, and at what rate

Average effective tax rates by net-wealth percentile, 2022 — their Figure 3



From percentile 67



From percentile 98.5

Current IP (red): flat zero until ~p99.3. Norway starts near p80; Piketty and Sim. NO climb steepest at the top.

Result 2 · Base breadth, not rate steepness, drives everything

Individual net wealth, 2022 — condensed from their Table 3 (p.21)

Design	Taxpayers	Revenue (€M)	Avg. rate t	Kakwani K	Suits S	Wealth Gini cut
IP (baseline)	170,404	1,126	0.02%	0.231	0.806	0.000%
Sim. NO	7,647,459	15,456	0.29%	0.191	0.522	0.066%
Sim. CH	12,400,000	4,843	0.09%	0.120	0.276	0.013%
Sim. FR	117,585	1,444	0.03%	0.244	0.893	0.013%
Sim. 1	1,834,488	3,709	0.07%	0.226	0.752	0.013%
Sim. 2	201,687	2,405	0.04%	0.234	0.849	0.013%
Sim. 3	4,626,929	12,673	0.24%	0.231	0.779	0.066%
Sim. Piketty	11,010,000	15,861	0.30%	0.197	0.570	0.080%

Revenue spans 14× — €1.1bn (IP) to €15.9bn (Piketty).

Formal progressivity falls as bases broaden: K and S are highest for IP/FR/Sim. 2.

Short-run Gini moves are tiny for wealth — the action is in revenue and the long run.

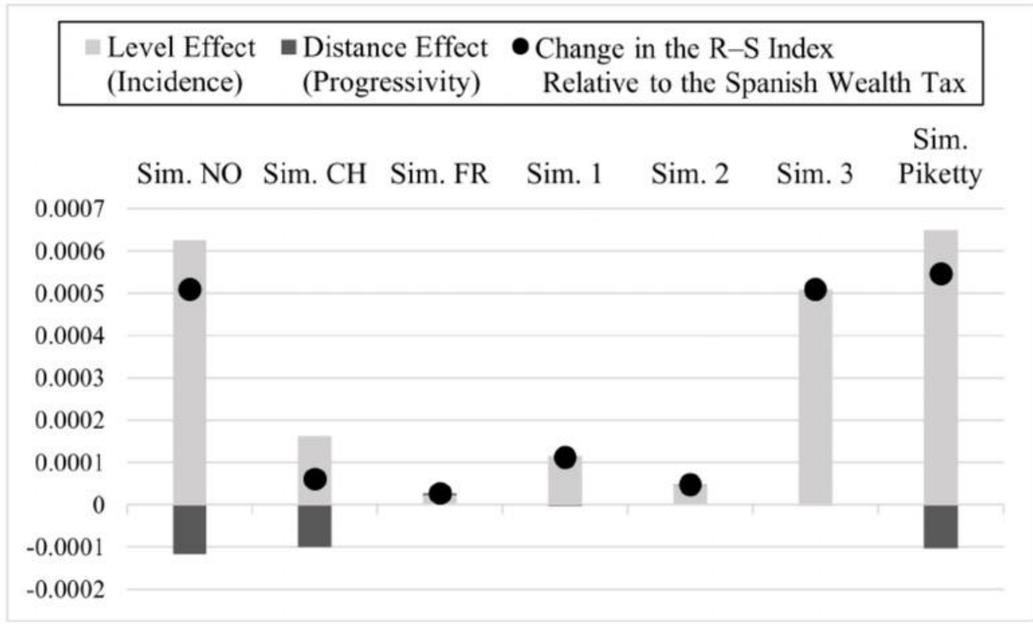
Result 3 · Redistribution comes from incidence, not rate design

Change in Reynolds–Smolensky vs the current IP, split into level and distance effects — their Figure 4

Every alternative redistributes more than the current IP — and in every case the gain is **dominated by the level effect**: the tax simply collects more.

The distance (structural-progressivity) effect is small and often **negative** for the broad-base designs.

> Their own words: improvements are “driven primarily by the Level Effect” — incidence, not progressivity (p.22).



ΔR–S vs the Spanish IP, individual net wealth

Result 4 · What the revenue could buy: a universal child benefit

Each design's revenue recycled as an equal per-child transfer — their Table 4 (p.25)

Design	€ / child / year	Disposable-income Gini cut
IP (baseline)	149	0.48%
Sim. NO	2,049	4.58%
Sim. CH	642	1.40%
Sim. FR	191	0.56%
Sim. 1	492	1.19%
Sim. 2	319	1.06%
Sim. 3	1,680	4.21%
Sim. Piketty	2,103	4.87%

The current IP finances **€149 per child per year**. Broad-base designs finance **€1,680–2,103**.

Recycling the revenue cuts the disposable-income Gini by **4.2–4.9% (1.6–1.8 Gini points)** under Sim. 3 / Piketty — versus 0.5% under the current tax.

Redistribution tracks revenue capacity, not the shape of the rate schedule.

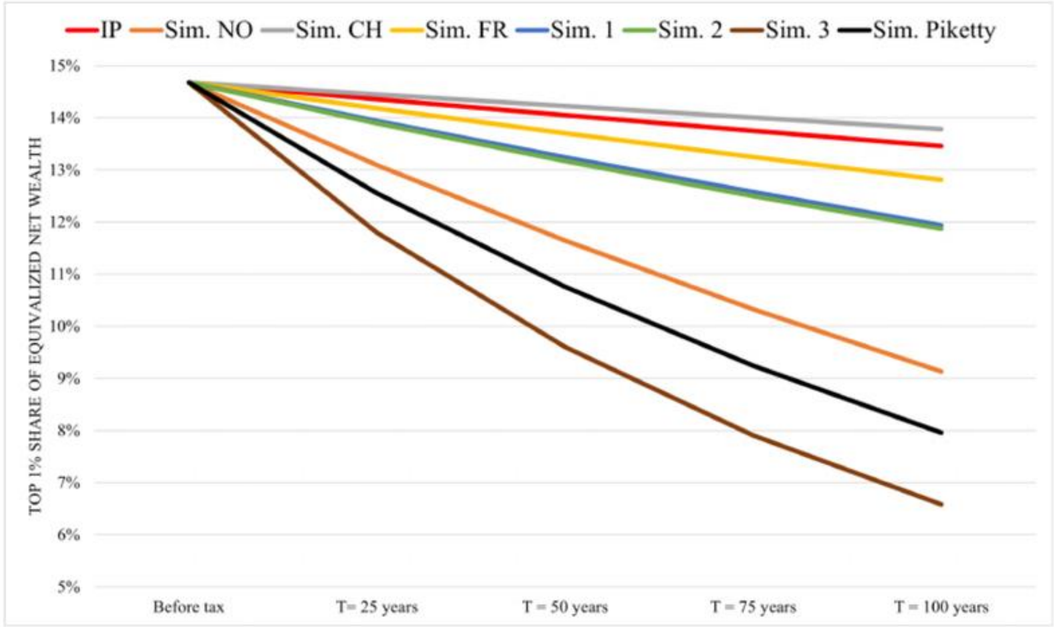
Result 5 · Compounding turns small rates into large long-run wedges

Top 1% share of equivalized net wealth under repeated annual taxation, g = 2% — their Figure 5

Current IP: top 1% share drifts from 14.7% to **13.5% after 100 years** (−1.2pp).

Norway-style and Piketty schedules: **below 8–9% by T=100** (−5.5 to −6.7pp); Sim. 3 falls furthest among Spanish options (to 6.6%, −8.1pp).

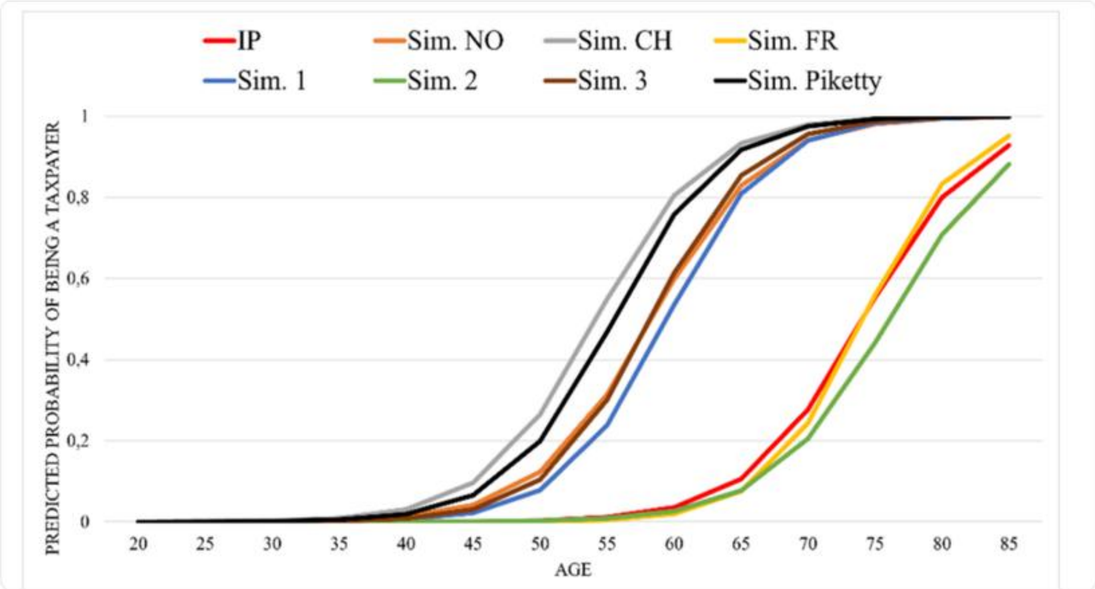
Mechanical projection, not a forecast — the authors present it as illustrating how annual stock taxation cumulates (p.25). Effective long-run rates reach ~22% of wealth for the broad designs (Table 5).



Top 1% share of equivalized net wealth, T = 0 → 100 years

Result 6 · Design choices reshape who pays

Taxpayer profiles across designs — their Figures 6–7 and Table 6



Predicted probability of paying, by age (logit with region and household controls)

Under the current IP and the French design, payment is an **old-age phenomenon** — probabilities take off only after ~65.

Broad bases (Norway, Switzerland, Sim. 1/3, Piketty) pull the curve left: **younger and more diverse payers** enter the base.

Women are less likely to pay under every design — the administrative footprint of the gender wealth gap.

Their conclusions

As stated in the paper (p.33–34) and their own deck

Current design (IP)

High formal progressivity, minimal effect: 0.4% of individuals pay, €1.1bn raised, no measurable inequality reduction; regional rebates create horizontal inequity.

Reformed designs (Sim. 3, Piketty, NO)

Broader bases and harmonized rules multiply taxpayers (4.6–11M) and revenue (€12.7–15.9bn), cut disposable-income inequality once recycled, and bend long-run wealth concentration.

> The policy claim: **redistributive gains come from broadening the base and raising effective incidence — not from steepening marginal rates.**

Discussion

Four comments from a tax-microsimulation seat

1 · The baseline is pre-ITSGF law — Sim. 2 is partly the status quo

Ley 38/2022 entered into force 29 Dec 2022 and accrues each 31 December — its first tax year is 2022 (your valuation date), not 2023 (fn. 2)

€623M
ITSGF collected on 2022 wealth (12,010 taxpayers; 89% Madrid) — ≈ half of Sim. 2's incremental €1,280M

€1,926M
actual IP receipts for 2023 (+€676M, 90.6% from Madrid ending its rebate under Ley 12/2023) — vs €1,126M simulated "current law"

−60%
ITSGF outturn vs its ~€1.5bn official forecast — a live lesson in static revenue scoring

None of this invalidates the design comparison — but the baseline should be labeled a **pre-ITSGF counterfactual**, and footnote 2's dating corrected.

> **The opportunity:** 2023 is now observable — Madrid un-rebated plus a state surcharge is a partial dress rehearsal of Sim. 2. An out-of-sample validation against AEAT's 2023 statistics would be a genuinely novel contribution.

2 · Static revenue is a ceiling — and your own citations price the gap

Behavioral erosion scales with the tax rate, so it is not ranking-neutral across the eight designs

A transparent ready-reckoner (illustration, not an estimate)

- Advani & Tarrant (2021): base −7–17% per 1pp → **Sim. NO €15.5bn → €12.8–14.4bn**
- Mas-Montserrat et al. (2025, your citation): −3.21% per 0.1pp ATR → ~8–10% off Sim. NO/3/Piketty
- Brülhart et al. (2022, your citation): ≥43% reported-wealth response per 1pp = the upper bound
- Saez–Zucman score with 15% (strong) / 50% (weak enforcement) erosion — zero is outside the published debate

The missing citation helps you

Agrawal, Foremny & Martínez-Toledano (AEJ: Applied 2025): Madrid's zero rate; mobility costs 6× more in income tax than it saves in wealth tax; **Madrid itself would gain from a harmonized positive rate** — your Sim. 2/3 case, made stronger.

Two anchors worth one sentence each

Norway's own tax yields ~0.5% of GDP vs Sim. NO's 1.13% of Spain's; and Sims 1–3 keep the 60% IRPF–IP cap — the channel carrying 92.6% of measured Spanish avoidance.

Because erosion rises with rates, adjusting for it **strengthens your breadth-over-rates conclusion** while shrinking the headline euros — worth owning explicitly.

3 · Two mechanical results need a re-run

Fixable before the journal version — neither threatens the qualitative ranking

The equivalized long-run Gini cannot rise

Under uniform growth the no-tax distribution is scale-invariant, and with your C_t = 0.974 the cumulative IP levy (0.50% by T=25) can move the Gini at most **~0.17pt, downward**. Table 5 reports **+1.04pt** — six times the bound, wrong sign; five designs stay above the starting Gini for a century. The projection code needs an audit.

Which total wealth? €5.3T vs €7.2T

Every printed t and $t_{s/r}$ coheres on total net wealth ≈ €5.3T (the EFF-consistent figure), but Table A.1's mean × count implies €7.2T — a 35% gap. Since $t_{s/r}$ seeds the entire long-run analysis, A.1's descriptives need reconciling.

Smaller items for the same consistency pass

Table 3's net-wealth R–S column is silently ×1000 vs Table A.4 · overlap matrix has impossible cells (Piketty∩Sim.2 = 297,219 > Sim.2's 201,687) · ratio 3.0838 vs 3.1125 on one page · seven vs eight vs nine scenarios · declared vs ×3.08 market values across scenarios (Zurich's housing base ≈ 2.2× the IP's) — state each design's valuation base explicitly.

4 · Say the finding your tables prove — and open the engine

Redistribution and measured progressivity move in opposite directions; replication is one artifact away

Framing

The abstract promises designs that "increase tax progressivity" — but Kakwani falls under every broad-base design you champion (0.231 → 0.191 Norway, 0.197 Piketty), and your own decomposition credits the level effect. The inverse trade-off *is* the interesting result: own it in the abstract. (Also: your baseline validates within 8–10% of AEAT's €1,250M — report that; it's a strength.)

Replicability

The Panel de Hogares is a joint INE–AEAT–IEF product any researcher can request free under the IEF protocol — unusually open for register data. What's missing is yours, not theirs: no code, no parameter tables (the Zurich design never states its schedule — statutory 0.3% top rate vs ~0.65% with multipliers). EUROMOD cannot simulate the IP at all ("lack of data on wealth").

The ask: publish the 17-region + 4-foreign-design parameter tables and the calculator code with the journal version. **We've started an open re-encoding of all eight designs** (github.com/MaxGhenis/iariw-2026-canto-garcia-replication) — with your parameters it becomes a full replication on any PH or EFF extract.

Questions to open the floor

Valuation base. Which real-estate value enters each scenario's taxable base — declared or ×3.08 market? What does Piketty raise on the IP's own base?

The 2023 test. Madrid un-rebated + ITSGF is a partial dress rehearsal of Sim. 2 — does AEAT's 2023 outturn (€1.93bn) match what your model predicts?

The cap. Is the 60% IRPF–IP limit active in Sims 1–3 and Piketty, and how much revenue does it remove from each?

Thank you — and over to **Olga Cantó** and **Francisco García-Rodríguez**

Max Ghenis · max@policyengine.org · review + open re-encoding: github.com/MaxGhenis/iariw-2026-canto-garcia-replication